Day and Night: Sky Shift

I can explain that Earth's rotation causes day and night and makes the Sun appear to move across our sky.

A fixed Sun

use evidence

A rotating Earth

use evidence

One place moving into and out of light

use evidence

Open the original interactive lesson

What success looks like

- Identify the lit side of Earth as day and the unlit side as night.
- Sequence one place on Earth through a full rotation.
- Use rotate and apparent to explain what an observer sees.

Retrieve and reconnect

- Last lesson: why can the Moon look different even though it stays spherical?
- Last week: state one way Earth moves in space.
- Older learning: why should one factor change at a time in a fair comparison?
- Lead-in: what might cause one place to move from day into night?

At sunset, what do you predict is really moving? Commit to a first idea; it is not marked.

Think first. Point, say, sign or write your choice. Give one reason.

A. The Sun travels around Earth once each day.

B. Earth turns while the Sun stays in the model's fixed position.

C. Clouds carry daylight away.

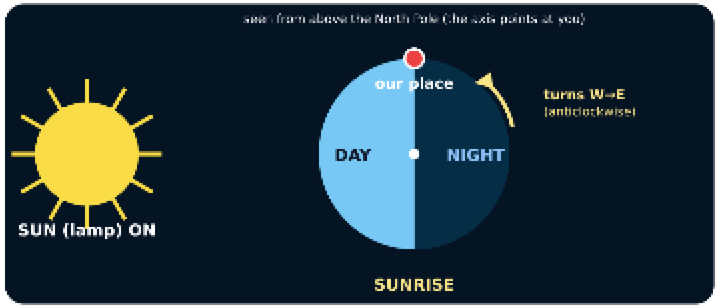
Look closely · what is the evidence?

Observation 1: the marked place is beside the light-dark boundary.

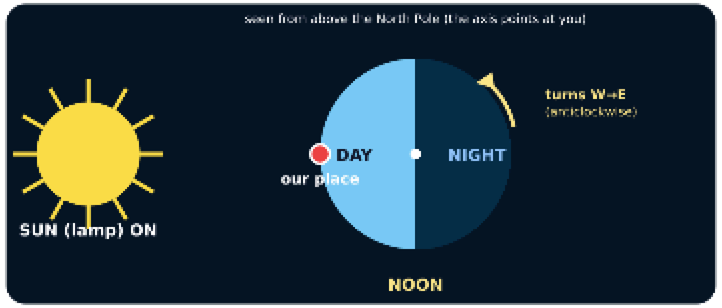
Observation 2: the same place is near the centre of the lit half.

Observation 3: the lamp is still on and the same place is on the unlit half.

Look closely · what is the evidence?



Earth beside a lamp; the marked place is at the edge of the lit half



The marked place is facing the lamp

The marked place is on the dark half; the lamp is still on

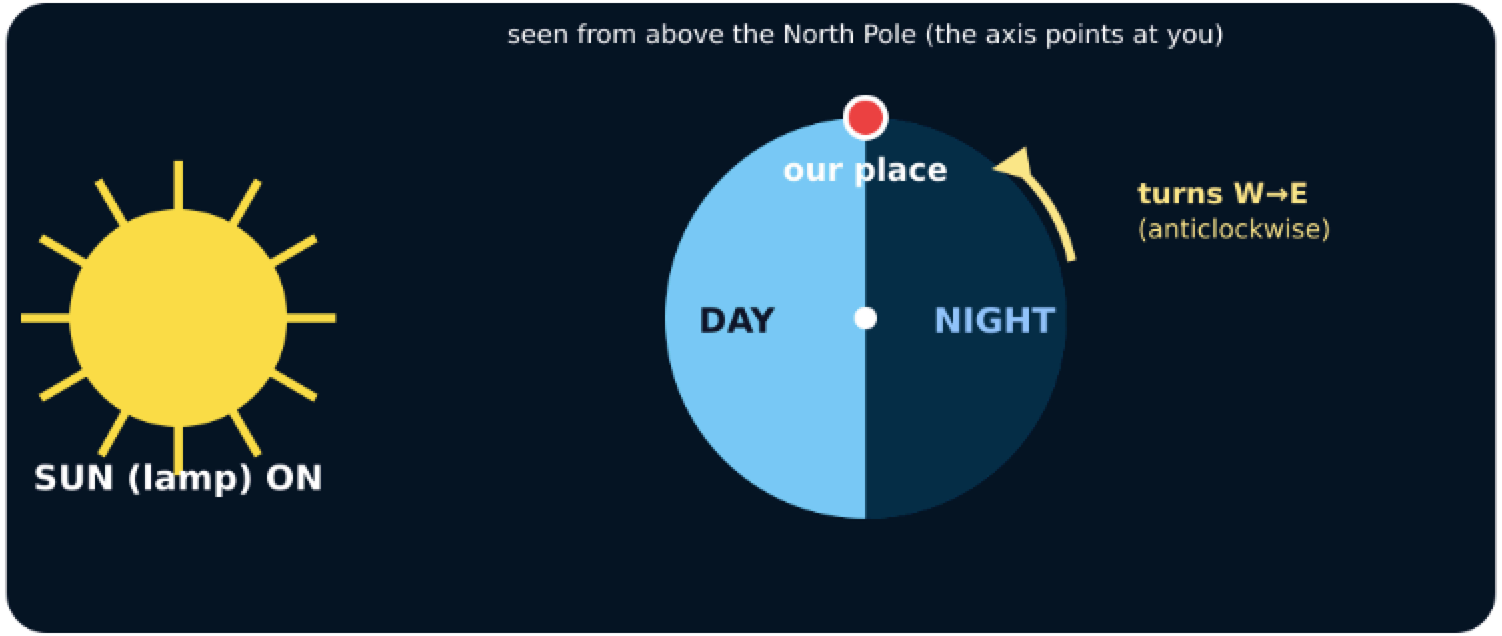
Words we will use

- rotate — turn
- axis — turning line
- apparent — seems to

Watch the model

- Fix a cool lamp as the Sun. Put one visible marker on a globe as our place.
- Notice that only the half facing the lamp is lit. Label that half day and the other half night.
- Turn the globe west to east on its axis. Track the marker as it enters and leaves the light.

Watch the model



Rotating Earth model with a lamp as the Sun

The marked place is on the dark side and rotates towards the lamp. What comes next there?

A. Morning, because the place is entering the lit half.

B. Night, because the lamp has switched off.

C. A season, because one turn takes a year.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Stand at the marked place in the model and describe the lamp's position from that viewpoint.
- As Earth rotates west to east, the observer turns with it. The Sun therefore appears to move east to west: it seems to rise, cross the sky and set.
- Say the full link: Earth rotates west to east, so day and night repeat and the Sun's east-to-west daily movement is apparent.

Find and repair the misconception

Model glitch: the Sun travels around Earth once every day.

A. Keep it: a daily Sun orbit explains sunrise.

B. Repair it: Earth rotates, so the Sun only appears to cross our sky.

C. Repair it: clouds move the Sun from east to west.

Four-position sky lab

Match each observation at the marked place to its position in one rotation.

- Choose one evidence card.
- Commit it to one sky position.
- Use the feedback to repair any mismatch.
- Freeze at least two results, then explain the pattern.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- Sunrise
- Middle of the day
- Sunset
- Night

Use the frozen evidence to say what moved, what stayed fixed and why the Sun appeared to move.

Your independent mission

Create a four-position explanation of day, night and the Sun's apparent movement for one marked place.

Record: One saved or paper sequence with four positions, rotation arrows and a because sentence using apparent.

Choose your support and challenge

- Supported: Order four picture cards and complete: When our place faces the Sun it is __. When it faces away it is __.
- Standard: Draw and label four positions in one rotation, then explain day, night, sunrise and sunset.
- Stretch: Explain the same four positions to someone who thinks the Sun orbits Earth each day.

Show what you know

Explain why day and night happen and why the Sun appears to move across our sky.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.